VLSI PDA HW1

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I. Explain how the adjustments of the clock period and the core utilization affect the metrics (DRC violations, slack, chip area, and wire length).

1. Different configurations

Clock Period	Core Util	DRC viol	Slack	Chip Area	Wire Length
1800	0.5	1	-30.9	41107	223286
1850	0.6	1	-0.2	34640	202923
	0.58	0	0.5	35775	203200
	0.57	0	0	36348	207479
	0.55	0	0	37595	206496
	0.5	0	-0.1	41107	208861
	0.4	1	-0.8	50689	211352
1900	0.6	1	0	34640	199385
	0.58	3	0	35775	200123
	0.57	0	1.3	36348	202087
	0.56	0	0.5	36956	202171
	0.55	0	0.1	37595	203962
	0.5	0	0.9	41107	202823
2000	0.5	1	3.1	41107	198533
	0.4	1	2.6	50689	201849
	0.35	1	0.7	57484	196198

2. Core Utilization

The core utilization factor affect chip area directly in a inverse relationship. Also, higher core utilization generally leads to a reduction in wire length. However, I cannot observe a clear relationship between core utilization and DRC violation. In my experimental results, both too high and low utilization values tend to cause DRC violations.

3. Clock Period

Clock Period	Core Util	DRC viol	Slack	Chip Area	Wire Length
1800	0.5	1	-30.9	41107	223286
1850	0.5	0	-0.1	41107	208861
1900	0.5	0	0.9	41107	202823
2000	0.5	1	3.1	41107	198533

The clock period factor also directly impacts the slack value. The smaller clock period will get a smaller slack. A larger clock period provides more flexibility for routing optimization, which can contribute to shorter wire lengths.

II. Explain the purpose of inserting well tap cell

If the wells are not properly tied to VDD and VSS, it could cause a latch-up problem. The inserting of the well tap cells provide a low-resistance electrical connection from VDD to the N-well and from VSS to the P-well to prevent Latch-up.

III. Show the configuration from your best result.

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--- Summary of the Current Result ---
 Slack Time:          0.500
 DRC Violations:      0
 Clock Period:        1850.000
 Total area of chip: 35775.323
 Total wire length:  203200.4320
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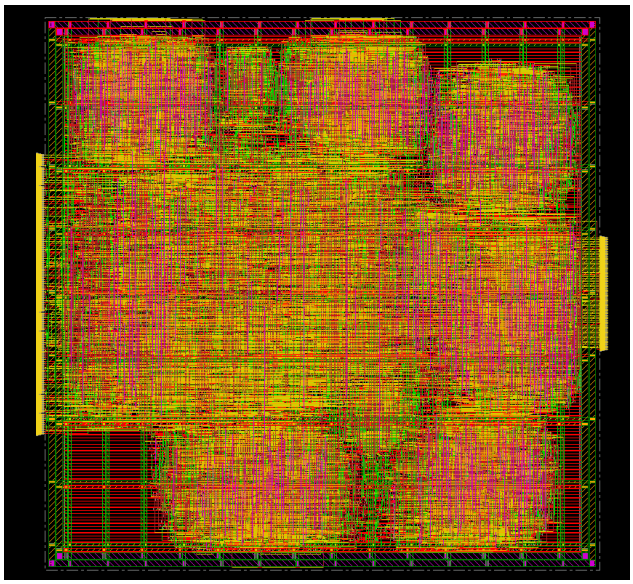
IV. Show the final chip layout